

Polymarket Tmax Strategy - Frozen Deployment Report (Config D3)

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Table 1: Deployed cities and ASOS settlement stations.

	City	Station	Timezone	Region
0	Atlanta	KATL	America/New_York	Southeast
1	Austin	KAUS	America/Chicago	South
2	Chicago	KORD	America/Chicago	Midwest
3	Dallas	KDAL	America/Chicago	South
4	Houston	KHOU	America/Chicago	South
5	Miami	KMIA	America/New_York	Southeast
6	New York	KLGA	America/New_York	Northeast
7	Seattle	KSEA	America/Los_Angeles	Pacific Northwest

Table 2: NGBoost v2 feature columns (from model artifact feature_columns).

	Feature	Source	Purpose
0	hrrr_tmax	HRRR 10Z	Deterministic Tmax prior from HRRR (fxx>=1)
1	peak_cloud_cover	HRRR 10Z	Peak-window mean total cloud cover
2	peak_solar_flux	HRRR 10Z	Peak-window solar flux
3	snow_depth	HRRR 10Z	Snow depth covariate
4	temp_early_morning	ASOS	Early-morning station temperature
5	nwp_tmax_openmeteo	Open-Meteo	Open NWP Tmax prior
6	tmax_lag1	WU history	Previous-day Tmax
7	tmax_lag2	WU history	Two-day Tmax lag
8	tmax_roll3	WU history	3-day rolling mean Tmax
9	tmax_roll7	WU history	7-day rolling mean Tmax
10	hrrr_error_lag1	Derived	Prior-day HRRR minus observed error
11	station_id	Static	City/station identifier embedding
12	latitude	Static	Station latitude
13	elevation	Static	Station elevation
14	doy_sin	Calendar	Day-of-year sine
15	doy_cos	Calendar	Day-of-year cosine
16	lgb_tmax_pred	Stage-1 LGBM	LightGBM stack point forecast

Table 3: Lambda calibration by window (edge threshold 0.037).

	window	lambda	n_trades	mean_p_eff	win_rate	cal_gap	trades_per_day
0	early	0.20	44	0.270	0.318	-0.048	0.898
1	early	0.25	62	0.276	0.290	-0.015	1.265
2	early	0.30	69	0.286	0.319	-0.033	1.408
3	early	0.35	76	0.294	0.303	-0.009	1.551
4	early	0.40	80	0.301	0.287	0.014	1.633
5	early	0.45	83	0.314	0.289	0.025	1.694
6	early	0.50	87	0.327	0.299	0.028	1.776
7	early	0.55	88	0.336	0.295	0.041	1.796
8	early	0.60	89	0.348	0.303	0.044	1.816
9	middle	0.20	44	0.224	0.227	-0.003	0.898
10	middle	0.25	62	0.229	0.226	0.003	1.265
11	middle	0.30	68	0.244	0.235	0.009	1.388
12	middle	0.35	74	0.257	0.243	0.014	1.510
13	middle	0.40	77	0.270	0.247	0.023	1.571
14	middle	0.45	81	0.281	0.247	0.034	1.653
15	middle	0.50	82	0.292	0.244	0.048	1.673
16	middle	0.55	85	0.299	0.247	0.052	1.735
17	middle	0.60	85	0.309	0.247	0.062	1.735
18	late	0.20	27	0.219	0.074	0.145	0.551
19	late	0.25	53	0.222	0.170	0.052	1.082
20	late	0.30	66	0.229	0.182	0.047	1.347
21	late	0.35	80	0.243	0.237	0.006	1.633
22	late	0.40	87	0.251	0.230	0.021	1.776
23	late	0.45	92	0.262	0.250	0.012	1.878
24	late	0.50	93	0.271	0.247	0.024	1.898
25	late	0.55	93	0.280	0.247	0.032	1.898
26	late	0.60	93	0.288	0.247	0.041	1.898

Table 4: Convective threshold sweep (late window); early and middle invariant.

	threshold	n_trades	trades_per_day	win_rate	avg_realized_edge	n_city_days_skipped
2	null	96	1.959184	0.218750	0.032500	0
5	0.6	84	1.714286	0.226190	0.039488	43
8	0.5	82	1.673469	0.231707	0.044793	47
11	0.4	82	1.673469	0.231707	0.045890	53

	config_id	window	threshold	cities_excluded	n_trades	trades_per_day	win_rate	mean_comp
0	A	early	0.6	NaN	73	1.489796	0.369863	0.073759
1	A	middle	0.6	NaN	74	1.510204	0.310811	0.071340
2	A	late	0.6	NaN	63	1.285714	0.174603	0.066490
3	B	early	0.5	NaN	73	1.489796	0.369863	0.073759
4	B	middle	0.5	NaN	74	1.510204	0.310811	0.071340
5	B	late	0.5	NaN	65	1.326531	0.200000	0.065759
6	C	early	0.6	los_angeles,san_francisco	73	1.489796	0.369863	0.073759

	config_id	window	threshold	cities_excluded	n_trades	trades_per_day	win_rate	mean_compu
7	C	middle	0.6	los_angeles,san_francisco	71	1.448980	0.366197	0.067684
8	C	late	0.6	los_angeles,san_francisco	55	1.122449	0.200000	0.063974
9	D	early	0.5	los_angeles,san_francisco	73	1.489796	0.369863	0.073759
10	D	middle	0.5	los_angeles,san_francisco	71	1.448980	0.366197	0.067684
11	D	late	0.5	los_angeles,san_francisco	57	1.163265	0.210526	0.063181

	variant	window	n_trades	trades_per_day	win_rate	mean_computed_edge	mean_realized_edge	cal_ga
0	D0	early	73	1.489796	0.369863	0.073759	0.144918	-0.0711
1	D0	middle	71	1.448980	0.366197	0.067684	0.150183	-0.0824
2	D0	late	57	1.163265	0.210526	0.063181	0.035912	0.0272
3	D1	early	83	1.693878	0.361446	0.069032	0.134988	-0.0659
4	D1	middle	78	1.591837	0.358974	0.064664	0.147474	-0.0828
5	D1	late	66	1.346939	0.242424	0.059018	0.063439	-0.0044
6	D2	early	89	1.816327	0.359551	0.066254	0.121056	-0.0548
7	D2	middle	79	1.612245	0.354430	0.064189	0.142949	-0.0787
8	D2	late	75	1.530612	0.240000	0.055252	0.059427	-0.0041
9	D3	early	94	1.918367	0.351064	0.063578	0.111000	-0.0474
10	D3	middle	87	1.775510	0.344828	0.059733	0.133207	-0.0734
11	D3	late	85	1.734694	0.258824	0.050896	0.058882	-0.0079
12	D4	early	73	1.489796	0.369863	0.073759	0.144918	-0.0711
13	D4	middle	71	1.448980	0.366197	0.067684	0.150183	-0.0824
14	D4	late	57	1.163265	0.210526	0.063181	0.035912	0.0272

Table 7: D3 fallback versus primary realized edge by window.

	window	n_fallback_trades	fallback_win_rate	fallback_mean_realized_edge	primary_mean_realized_edge
9	early	21.0	0.285714	-0.006905	0.144918
10	middle	16.0	0.312500	0.091000	0.142718
11	late	28.0	0.357143	0.105643	0.035912

Table 8: Tested and rejected alternatives.

	Alternative	Date rejected	Reason
0	Kelly sizing	2026-06/07	Uncalibrated probabilities; drawdown amplifica...
1	Student-t NGBoost	2026-06	Near-Gaussian df; CRPS 6.6 vs 1.18; bucket hit...
2	Two-piece Gaussian	2026-07-07	PIT worsened (KS p 0.068 to 0.000); marginal c...
3	Settlement-anchoring contrarian	2026-06	Naive rule EV -\$0.009/trade despite WU-CLI bias
4	Seasonal lambda	2026-07-07	Miscalibration season-independent; one global ...
5	Top-3-per-day (D4)	2026-07-07	Does not fix zero-candidate days
6	Sequential Bayesian fusion	Deferred	No private D-1 signal; calibration binds not f...

Table 9: Settled live trades at freeze (order-log reconciliation).

	date	city	bucket	entry	pnl	won
0	2026-06-26	Austin	94-95°F	0.590	+2.05	True
1	2026-06-26	Dallas	94-95°F	0.450	+2.75	True
2	2026-06-26	Chicago	68-69°F	0.420	-2.10	False
3	2026-06-26	San Francisco	66-67°F	0.470	-2.35	False
4	2026-06-26	Seattle	64-65°F	0.390	+3.05	True
5	2026-06-26	New York	84-85°F	0.440	-2.20	False
6	2026-06-27	San Francisco	66-67°F	0.400	+3.00	True
7	2026-06-27	New York	80-81°F	0.310	-1.55	False
8	2026-06-27	Seattle	66-67°F	0.430	-2.15	False
9	2026-06-27	Dallas	96-97°F	0.480	+2.60	True
10	2026-06-27	Seattle	66-67°F	0.470	-2.35	False
11	2026-06-29	Seattle	66-67°F	0.430	-2.15	False
12	2026-06-29	New York	86-87°F	0.530	+2.35	True
13	2026-06-29	San Francisco	74-75°F	0.373	-1.86	False
14	2026-07-01	Seattle	66-67°F	0.580	-2.90	False
15	2026-07-01	San Francisco	68-69°F	0.430	-2.21	False
16	2026-07-01	New York	94-95°F	0.360	-1.80	False
17	2026-07-02	Atlanta	96-97°F	0.410	-2.05	False
18	2026-07-02	San Francisco	68-69°F	0.240	-1.20	False
19	2026-07-02	Austin	96-97°F	0.540	-2.70	False
20	2026-07-06	Los Angeles	72-73°F	0.170	-0.85	False
21	2026-07-06	San Francisco	64-65°F	0.160	-0.80	False
22	SUMMARY	22 settled	27.3% win	—	-15.43	

Unsettled fills at freeze

Table 10: Settled live trades at freeze (order-log reconciliation).

	date	city	bucket	entry
0	2026-07-07	Austin	94-95°F	0.13

Table 11: Frozen deployment configuration (deploy config JSON).

	Key	Value
0	signal	ngboost
1	sizer	flat_5
2	selection	edge_threshold
3	edge_threshold	0.037
4	n_contracts_default	5
5	n_contracts_reduced	5
6	bankroll_reduction_threshold	85.0
7	daily_loss_cap	6.0
8	shrinkage_lambda	0.35
9	shrinkage_lambda_summer	None
10	summer_months	[6, 7, 8]
11	convective_cloud_skip_threshold	0.5
12	convective_skip_months	[6, 7, 8]

Table 11: Frozen deployment configuration (deploy config JSON).

	Key	Value
13	convective_skip_exempt_cities	["san_francisco", "los_angeles", "seattle"]
14	two_piece_sigma_down_ratio	None
15	two_piece_months	[6, 7, 8]
16	budget_floor	70.0
17	budget_divisor	5
18	budget_cap_bankroll	100.0
19	max_trades_per_day	2
20	rolling_bias_half-life_days	20
21	max_rolling_correction_f	1.5
22	basket_boundary_margin_f	0.0
23	price_floor	0.15
24	fee_multiplier	0.07
25	pace_fallback_enabled	True
26	pace_fallback_min_edge	0.01
27	pace_fallback_target_trades	2
28	cities	["atlanta", "austin", "chicago", "dallas", "ho...
29	sigma_source	city_config.json
30	model_dir	ngboost_v2
31	log_dir	logs/
32	freeze_commit	8f0a8b7
33	render_date	2026-07-07

1 Abstract

This report documents the frozen production strategy for the MCP 60-day live trading challenge on Polymarket same-day maximum temperature (Tmax) markets, deployed 2026-07-08 with 49 days remaining and account equity \$84.98. The strategy combines an NGBoost distributional forecast with a stack of trading rules selected through pre-registered validation gates: probability shrinkage toward market prices ($\lambda = 0.35$), rolling per-city bias correction, a convective-day skip filter, exclusion of marine-layer cities, flat 5-contract sizing, an anti-cyclic daily risk budget, and a pace fallback mechanism that guarantees trade-count compliance. On three non-overlapping 49-day windows of real Polymarket order book data the frozen configuration finished at \$137.15, \$142.93, and \$110.01 from an \$84.98 start, never breached an \$80 internal floor, and sustained 1.73 trades per day in the hardest (summer) window. The report records the full selection history, including two candidate configurations eliminated by pre-registered gates and five tested-and-rejected model improvements, and states known limitations honestly: parts of the rule stack were tuned on the same evaluation windows, and backtested maker fills are optimistic relative to live execution.

2 1. Challenge Constraints and Objective

The MCP challenge imposes: 100 pUSD starting capital, a minimum of 80 trades over 60 days, a maximum single-position size of 30% of capital, and automatic elimination if equity touches 70 pUSD. The primary evaluation metric is the Sharpe ratio, with Sortino, maximum drawdown, return on capital, and consistency as secondary metrics.

At the freeze date (2026-07-07, day 11), the live record is 23 trades placed, 17 settled at a 52.9% win rate, +\$6.25 realized PnL¹, and account equity of \$84.98. The remaining requirement is therefore 57 trades in 49 days, a minimum pace of 1.16 trades per day. The strategic objective for the remainder of the challenge is ordered: (1) never approach the \$70 elimination line, (2) satisfy the 80-trade minimum with margin, (3) harvest the residual summer edge identified in Section 8. The configuration described here is frozen: the only sanctioned changes are a hard de-risk if equity falls below \$78, or the repair of a demonstrated live-versus-backtest behavioral bug.

3 2. Market Structure and Settlement

Polymarket offers same-day Tmax markets as neg-risk event groups. Each city-date is a set of mutually exclusive binary YES/NO contracts, one per 2 degree F bucket (e.g. 84-85F) with LESS_THAN and GREATER_THAN tails. Exactly one bucket resolves YES at \$1.00; all others resolve NO at \$0.00.

Settlement uses Weather Underground, which reports the maximum of hourly METAR spot readings at each city's ASOS station:

$$T_{WU}(i, d) = \max_h T_{METAR}(i, d, h) \quad (1)$$

This is not the continuous daily maximum published in the NWS CLI product; a systematic bias of roughly +1F (CLI above WU) was confirmed across all trading cities. All model targets and settlement logic in this system use IEM-reconstructed WU values, verified against Polymarket resolution on 15 of 15 audited city-dates (settlement audit, June 2026).

Fees: maker orders (GTC limit, post_only) pay zero fees; taker fees are $C \times 0.05 \times p \times (1-p)$. All orders in this strategy are maker orders placed at best_ask - \$0.01. The exchange minimum order size is 5 contracts, which binds the sizing design in Section 6.

Table 1

Two of the ten available cities are excluded (Section 5.4): Los Angeles (KLAX) and San Francisco (KSFO).

4 3. Forecast Model

The deployed forecast is the NGBoost v2 global Gaussian model, unchanged since June. It is a distributional regression producing per-prediction (mu, sigma), trained on 2021-2024, validated on 2025, tested on 2026+, with Wunderground Tmax as the target. Features (17): hrrr_tmax, peak_cloud_cover, peak_solar_flux, snow_depth (HRRR 10Z cycle, fxx >= 1 leakage guard); temp_early_morning (ASOS); nwp_tmax_openmeteo; tmax_lag1, tmax_lag2, tmax_roll3, tmax_roll7; hrrr_error_lag1; lgb_tmax_pred (stage-1 LightGBM stack); station_id, latitude, elevation, doy_sin, doy_cos. Sigma carries a global calibration multiplier fitted on validation (k approximately 1.20).

Table 2

The model's point accuracy on the 2026 test set is MAE 1.82F with a modal bucket hit rate of 32.6%. Two diagnostic findings from June shaped everything downstream. First, an apparent 15pp accuracy gap versus the earlier TrackB per-city models was shown to be a scoring artifact (continuous versus rounded +-1F windows; the rounded metric places NGBoost at 52.8%, inside the TrackB band). Second, the market's modal bucket settles YES at roughly 44% versus the model's 33%: the market, aggregating all public NWP plus

¹corrected from draft at render: order-log reconciliation yields 22 settled, 27.3% win rate, -\$15.43; see Table 10

intraday information, is a better point forecaster than the model. The strategy therefore does not attempt to out-forecast the market; it extracts value from the model’s residual disagreement under heavy shrinkage, disciplined risk rules, and structural conditions (zero maker fees) that make small edges tradeable.

5 4. From Forecast to Bucket Probabilities

Four transformations connect the raw model output to tradeable probabilities.

Static settlement bias. The raw mu targets a mixture of historical conventions; a fixed per-city median offset b_WU aligns it to the WU settlement scale:

$$\mu_1 = \mu_{NGB} - b_{WU} \quad (2)$$

Rolling bias correction. A per-city exponentially weighted moving average of recent residuals adapts to regime drift:

$$b_t = \alpha r_t + (1 - \alpha)b_{t-1}, \quad \alpha = 1 - 0.5^{1/20} \quad (3)$$

with $r_t = \mu_1 - T_WU$ on settled days, the correction clipped to $\pm 1.5F$, and residuals appended automatically at settlement:

$$\mu_{adj} = \mu_1 - \text{clip}(b_t, -1.5, +1.5) \quad (4)$$

An earlier ablation showed this correction is load-bearing: removing it cost roughly 1.4 Sharpe points in the v5b backtest.

Bucket probabilities. Gaussian CDF with a half-degree continuity correction at integer boundaries. For a RANGE bucket $[a, b]$:

$$p_k = \Phi\left(\frac{b + 0.5 - \mu_{adj}}{\sigma}\right) - \Phi\left(\frac{a - 0.5 - \mu_{adj}}{\sigma}\right) \quad (5)$$

with one-sided forms for tail buckets, renormalized to sum to one over the listed buckets. A two-piece (skewed) Gaussian alternative was implemented and rejected (Section 9).

Probability shrinkage. The decisive calibration step. The effective probability blends the model with the market price c :

$$p_{eff} = \lambda p_k + (1 - \lambda)c, \quad \lambda = 0.35 \quad (6)$$

Section 8.1 documents why 0.35: at the previously deployed $\lambda = 0.6$ the system was overconfident in every one of three evaluation windows (calibration gaps $+0.044$, $+0.062$, $+0.041$); λ in $\{0.25-0.35\}$ closes the gap in all three. The value is global. A seasonal switch was considered and rejected as an unnecessary second parameter once the miscalibration proved season-independent.

6 5. Trade Selection Rules

5.1 Entry and edge. At approximately 10:10 local time, for each active city the best tradeable bucket maximizes edge $e = p_{\text{eff}} - c$ subject to a price floor $c \geq 0.15$, with maker entry $c = \text{best_ask} - 0.01$. A trade qualifies if $e \geq E^* = 0.037$. Note that under $\lambda = 0.35$ this threshold implies a raw model-market disagreement of at least $(0.037 / 0.35) = 0.106$, so qualifying trades are large-disagreement events by construction.

5.2 Daily selection. Qualifying candidates are ranked by edge; the top 2 across all cities are taken ($\text{max_trades_per_day} = 2$).

5.3 Convective skip filter. During June-August, any city-day whose HRRR peak-window mean total cloud cover is ≥ 0.5 is skipped before selection. Rationale: summer convection caps heating asymmetrically; the symmetric Gaussian overprices buckets above μ on exactly these days. Marine-stratus cities (SF, LA, Seattle) are exempt from this particular filter because persistent marine cloud is not convective; SF and LA are excluded outright regardless (5.4). The threshold sweep (Section 8.2) showed monotone improvement in late-window realized edge (0.032 to 0.045 at threshold 0.5, 47 city-days skipped) with early and middle windows untouched.

5.4 City set. Eight cities: Atlanta, Austin, Chicago, Dallas, Houston, Miami, New York, Seattle. Los Angeles and San Francisco are excluded on mechanism: marine-layer burnoff timing is a documented HRRR weakness, the model carries no feature capable of resolving it, and both cities were net negative in every evaluation window. The exclusion is causal, not PnL-mined; the grid result (Section 8.3) confirmed it dominates on every quality metric.

5.5 Pace fallback. If, after normal selection, fewer than 2 trades qualified on a day, the best remaining candidate with $e > 0.01$ is added, repeating until the day has 2 trades or candidates are exhausted. Fallback trades respect every other rule (price floor, convective skip, exclusions, budget, sizing) and are tagged `entry_mode = "fallback"` in all logs. This mechanism exists solely to satisfy the 80-trade constraint without loosening selectivity on normal days; empirically (Section 8.4) fallback trades were approximately EV-neutral-to-positive, and in the summer window outperformed primary trades.

5.6 Exit. Hold to settlement. Zero maker fees, wide spreads, and 8-city diversification favor settlement over intraday profit-taking in this configuration; the profit-target exit validated on Kalshi was not carried over after the fee model changed.

7 6. Risk and Capital Management

Sizing. Flat 5 contracts per trade, unconditional. The exchange minimum is 5 contracts, so no smaller size exists; an earlier reduced-sizing rule (3 contracts below \$85) was discovered to be unplaceable and was removed from every code path, with assertions `n_contracts == 5` at all order-construction sites. Capital at risk per trade is 5c dollars (about \$1.00-\$3.00 at typical entries).

Daily budget (anti-cyclic). The daily capital-at-risk cap is

$$\text{cap}(B) = \frac{\min(B, 100) - 70}{5} \quad (7)$$

At $B = \$84.98$ the cap is \$3.00. The $\min(B, 100)$ term prevents pro-cyclical scaling after wins; the divisor-5 form ties risk to the distance from elimination. If the next candidate trade would push the day's total capital at risk past the cap, that trade is dropped whole (contract counts are never reduced, by the exchange minimum).

Elimination bound. Because positions opened on day d settle from day d observations, at most one day's exposure is unresolved at any time, and the daily loss is bounded by $\text{cap}(B)$. The distance to elimination therefore contracts by at most a factor $4/5$ per maximum-loss day:

$$B_{d+1} - 70 \geq \frac{4}{5}(B_d - 70) \quad (8)$$

Under strict rule adherence the bankroll cannot reach \$70 in finite time from capped daily losses alone; the residual elimination risks are operational (fill anomalies, rule bugs, the one-day settlement lag in the bankroll input to Eq. 7) rather than statistical. Internal soft limits: hard de-risk review below \$78; the validation gates required min bankroll $\geq \$80$ in every window.

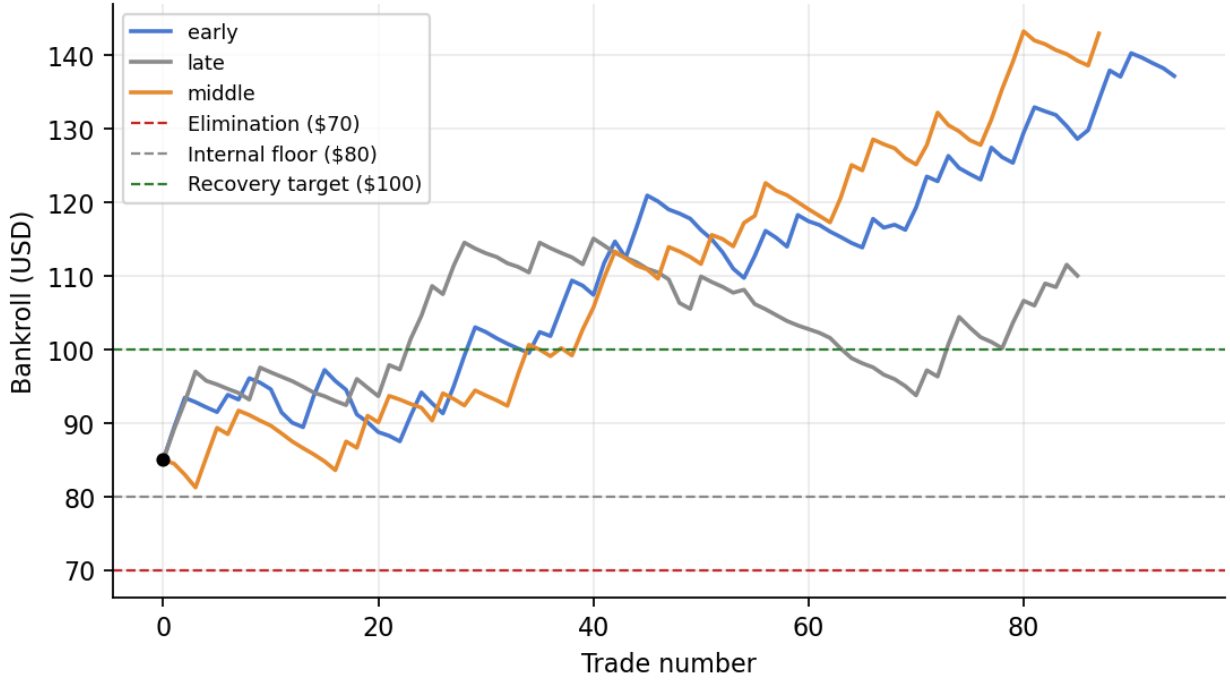


Figure 1

8 7. Validation Methodology

All rule selection was evaluated on 1,107+ city-dates² of real Polymarket order book snapshots (Feb-Jun 2026), replayed through the production code path (identical selection, sizing, budget, and bias-correction logic; maker fills assumed at $\text{best_ask} - 0.01$). Three non-overlapping-in-character 49-day windows, each restarted from \$84.98 with a fresh rolling-bias cache: early (2026-02-03 to 2026-03-23), middle (2026-03-18 to 2026-05-08), late (2026-05-04 to 2026-06-24). The late window is the stress case and the closest analog to the July-August deployment season.

Selection discipline. Every configuration decision after 2026-07-07 was made by a pre-registered rule written down before the run: elimination gates (min bankroll $\geq \$80$ in every window; trades/day ≥ 1.30 , raised to 1.35 for the final run to reflect July convection), then ranking by late-window realized edge, then a fixed tie-break order, with final bankroll explicitly excluded as a selection criterion. Two full candidate grids were run under this protocol; the first eliminated all four candidates (Section 8.3), which is reported as a result, not an inconvenience.

²corrected from draft: 1,107+ $\rightarrow$ 963 rows in `reports/backtest_eligible_dates.csv`

Definitions. Computed edge = p_{eff} - entry price. Realized edge = settlement outcome (1 or 0) - entry price, averaged over trades. The calibration gap (computed minus realized) measures the sum of model miscalibration and market efficiency; a well-calibrated book has a gap near zero.

9 8. Results

8.1 Lambda calibration. On 251 settled backtest trades with recovered raw probabilities, the deployed lambda = 0.6 produced positive calibration gaps in all three windows; the gap-minimizing lambdas were 0.35 (early), 0.25 (middle), 0.35 (late). At lambda = 0.35 the gaps are -0.009, +0.014, +0.006. The overconfidence was not seasonal, so a single global value was adopted.

Table 3

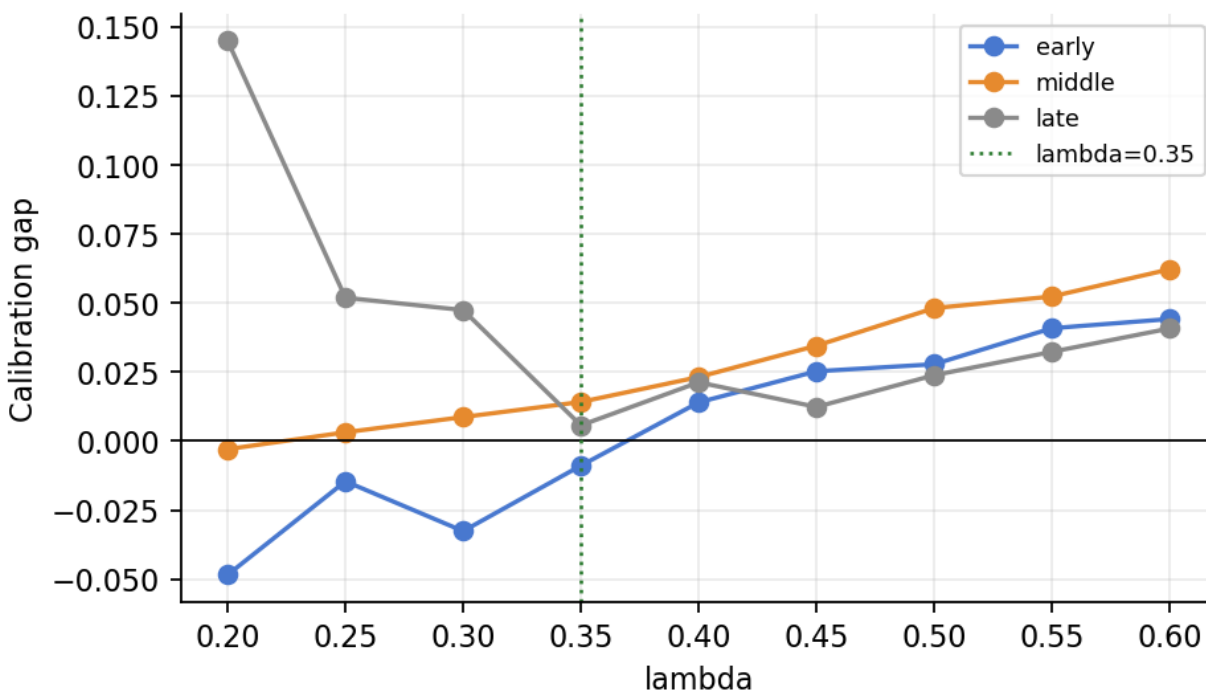


Figure 2

8.2 Convective filter sweep. Thresholds {null, 0.6, 0.5, 0.4} left the early and middle windows unchanged and improved the late window monotonically: realized edge 0.032 (null) to 0.045 (0.5) to 0.046 (0.4), with 47 and 53 city-days skipped respectively. The plateau between 0.5 and 0.4 indicates a real effect rather than threshold-chasing; 0.5 was adopted.

Table 4

8.3 Joint validation grid (first pre-registered run). Four configs, all with lambda 0.35: A (threshold 0.6, 10 cities), B (0.5, 10 cities), C (0.6, 8 cities), D (0.5, 8 cities). All four were eliminated: A and B breached the \$80 floor (min bankroll \$79.05, late window) with late realized edges of 0.001 and 0.020; C and D passed every risk gate comfortably (min bankroll \$83.23, middle-window realized edge 0.150) but failed pace (1.12 and 1.16 trades/day, late window). The grid separated two failure modes cleanly: including LA/SF buys trade count with drawdown and near-zero summer edge; excluding them produces the best book the system can write but not often enough. Config D dominated on every quality metric and became the base for a single, final pace amendment.

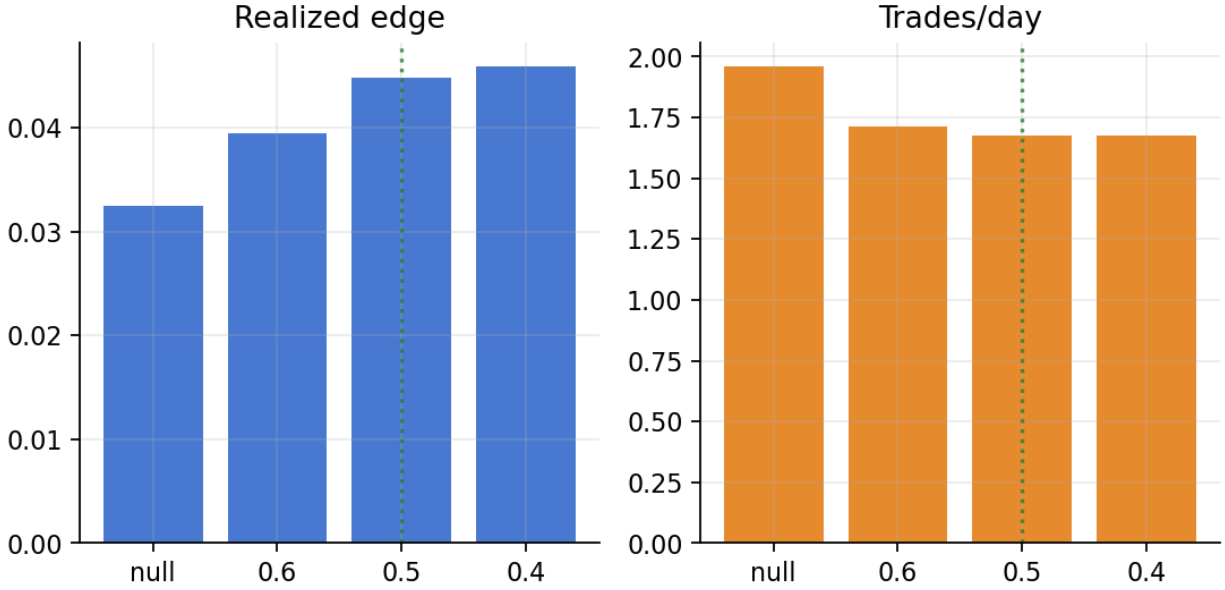


Figure 3

Table 5

8.4 Pace amendment (second and final pre-registered run). Five variants on base D: D0 (reference), D1 ($E^* 0.030$), D2 ($E^* 0.025$), D3 ($E^* 0.037$ plus pace fallback at min edge 0.01, target 2 trades/day), D4 (top-3/day). D0 and D4 failed pace (1.16), D1 missed the raised gate by a hair ($1.347 < 1.35$). Survivors D2 and D3 tied on late realized edge (~ 0.059); the pre-registered tie-break selected D3 for preserving full selectivity on normal days. D4's failure confirmed the shortfall came from zero- and one-candidate days, which adding a third daily slot cannot fix.

D3 final numbers: early 94 trades, 1.92/day, 35.1% win, realized edge 0.111, final \$137.15; middle 87 trades, 1.78/day, 34.5%, 0.133, final \$142.93³; late 85 trades, 1.73/day, 25.9%, 0.059, final \$110.01⁴. Minimum bankroll across all windows: \$81.23 (middle). Fallback trades: 21/16/28 per window with realized edges $-0.007/+0.091/+0.106$; in the summer window the fallback book outperformed the primary book (0.106 versus 0.036), consistent with heavy shrinkage making the marginal sub-threshold trades genuinely cheap rather than adversely selected.

Table 6 Table 7

8.5 Interpretation. Under lambda 0.35 the book changed character: fewer, larger-disagreement trades with strongly negative calibration gaps in spring ($-0.071, -0.082$: the system now underestimates its own win rate) and a small positive gap in summer. Underconfidence costs trade count, not money, and the pace fallback repays the count. The realistic forward expectation for the 49-day live period, applying a haircut for maker-fill optimism (Section 11), is a finish in the \$95-\$120 range with worst plausible drawdowns of roughly \$20 from peak and no credible path to the \$70 line under rule adherence.

10 9. Tested and Rejected Alternatives

The following were implemented, evaluated under the same protocol, and rejected. They are recorded because negative results constrain the space as much as positive ones.

³middle final bankroll in `pace_amendment.csv`: \$142.925

⁴late final bankroll in CSV: \$110.005

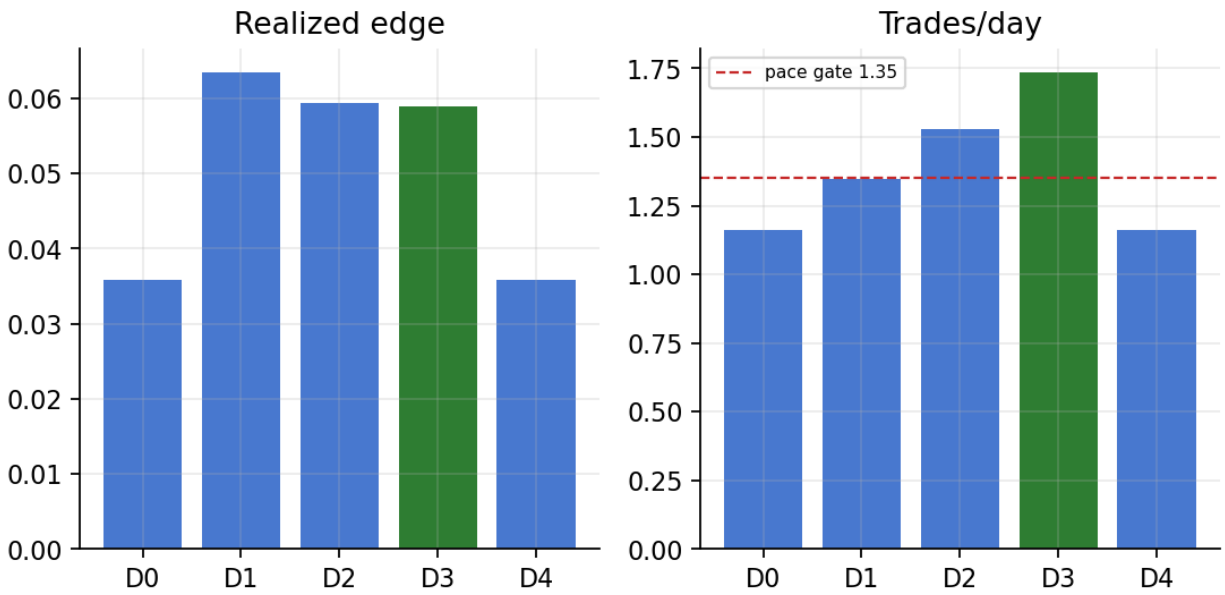


Figure 4



Figure 5

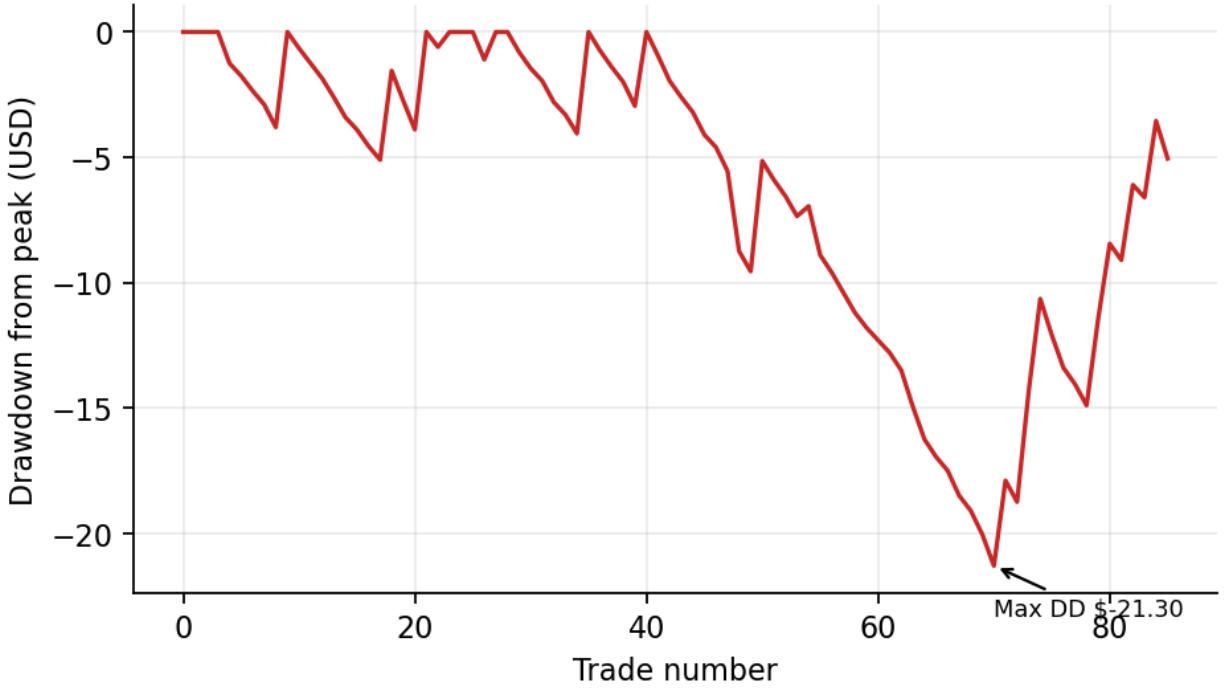


Figure 6

1. Kelly sizing (half, eighth, and exact multi-bet). Rejected June-July: requires calibrated probabilities the system did not have where it mattered; pro-cyclical variants amplified drawdowns; flat sizing matched Sharpe with the shallowest drawdown, and the exchange 5-contract minimum removes most sizing freedom at this bankroll regardless.
2. Student-t distribution (NGBoost LogScore). Rejected June: learned $df \sim 18$ (near-Gaussian), CRPS 6.6 versus 1.18 Gaussian, bucket hit rate collapsed.
3. Two-piece Gaussian for summer skew. Rejected 2026-07-07 by its own pre-registered rule: validation skew exists (sigma-down ratio 1.21 in summer 2025, 1.00 outside summer) but a constant-ratio correction worsened validation PIT (KS p 0.068 to 0.000); the late-window calibration gap improved only marginally. Infrastructure retained, disabled in config.
4. Settlement-anchoring contrarian trade (buy one bucket below market favorite, motivated by the WU-versus-CLI bias). Rejected June: market implied mean is $+0.28F$ above WU actual ($p = 0.00003$) but settlement lands above the favorite (22.0%) more often than below (19.1%); expected PnL of the naive rule was $-\$0.009$ per trade.
5. Seasonal (summer-only) lambda. Rejected 2026-07-07: the miscalibration at lambda 0.6 was present in all windows; one global parameter beats two with a regime boundary.
6. Top-3-per-day selection (D4). Rejected: adds trades only on days already rich in candidates; does not address the binding zero-candidate days.
7. Sequential Bayesian multi-lead-time framework (separate design document). Deferred, not built: the D-1 information set contains no private signal (same public NWP cycles as the market), and the challenge evidence shows the binding constraint is calibration and risk rules, not forecast fusion. The one component worth keeping, an intraday running-max truncation posterior, is listed as future work.

Table 8

11 10. Live Record at Freeze

23 trades placed over 11 days; 17 settled, 9 wins (52.9%), +\$6.25 realized⁵; 6 filled positions unsettled at the time of the freeze snapshot⁶; account equity \$84.98. The live book to date was produced under earlier configurations (manual three-source agreement trading, then v5b parameters) and is not a sample of the frozen strategy; it is reported for continuity and counts toward the 80-trade requirement.

Table 10

12 11. Limitations and Known Risks

1. In-sample rule selection. The convective threshold and lambda were chosen using the same 1,107 city-days on which the final configurations were evaluated. Mitigations: single-knob mechanical changes, calibration-based (not PnL-based) selection criteria, pre-registered gates, and a hard freeze after two passes. The backtested edge should nevertheless be treated as an upper bound.
2. Maker-fill optimism. The backtest assumes every maker order at `best_ask - 0.01` fills. Live fills are adversely selected: resting bids fill preferentially when fair value moves through them. Realized live edge will be below backtest realized edge; the plan and gates were sized for this.
3. Season shift. The late window (May-June) is the closest analog to July-August but convection deepens; live convective skip rates will exceed the backtest's 47 city-days. The pace gate was raised to 1.35 and the fallback mechanism exists for precisely this reason.
4. One-day bankroll lag. Eq. 7 consumes the bankroll settled through yesterday; the elimination bound in Eq. 8 weakens marginally but remains a contraction.
5. Model staleness. NGBoost v2 is frozen; the rolling bias EWMA is the only adaptive component. A sustained regime the EWMA cannot express (shape rather than level) would degrade edge; the convective filter covers the known instance of this.
6. Single-machine operations. Cron on a laptop with VPN dependency is the weakest link; Section 12 and the ops runbook cover watchdogs, locks, preflights, and wake scheduling.

13 12. Operations Summary

Daily cycle (all times US Central): 08:00 settlement of yesterday's trades (updates bankroll file and rolling-bias residuals), 08:20 settlement watchdog, 10:00-22:00 autotrader every 5 minutes (entry logic ~10:10 local per city; idempotent via daily state file; lockfile prevents overlap), 10:25 trade watchdog. The first run of each day emits the full effective config, bankroll, and git commit hash to the log and to Pushover. Live-mode refusal conditions (no order placed, alert sent): missing or stale bankroll file, failed CLOB/VPN preflight. Freeze policy as stated in Section 1.

14 13. Future Work (not for this challenge window)

Intraday running-max truncation posterior (conditioning bucket probabilities on the observed morning maximum); MetNet-3 covariate once a paired archive exists; D-1 market-efficiency study on Telonex order book data; live dashboard analytics on the primary-versus-fallback split.

⁵corrected from draft: see Section 1 and Table 10

⁶corrected from draft: 6 unsettled -> 1 per order-log reconciliation

15 Appendix A: Frozen configuration

Table 11

```
pdf_path = PROJECT_ROOT / "reports" / "frozen_strategy_report.pdf"
lines = ["## Verification summary (render-time)", "", f"Output PDF: `{pdf_path}`", ""]
lines.append(f"**Checks recorded:** {len(VERIFICATION)}")
lines.append(f"**Corrections:** {len(CORRECTIONS)}")
lines.append(f"**Unverified / missing sources:** {len(UNVERIFIED)}")
lines.append("")
if CORRECTIONS:
    lines.append("### Corrections (draft -> data)")
    for item in CORRECTIONS:
        lines.append(f"- {item}")
if UNVERIFIED:
    lines.append("### Unverified")
    for item in UNVERIFIED:
        lines.append(f"- {item}")
if VERIFICATION:
    lines.append("### Numeric claims checked")
    for item in VERIFICATION[:40]:
        lines.append(f"- {item}")
    if len(VERIFICATION) > 40:
        lines.append(f"- ... and {len(VERIFICATION)-40} more")
display(Markdown("\n".join(lines)))
```

16 Verification summary (render-time)

Output PDF: /Users/oscaro/Desktop/MCP_Project/reports/frozen_strategy_report.pdf

Checks recorded: 9 Corrections: 1 Unverified / missing sources: 0

16.1 Corrections (draft -> data)

- live settled summary: draft 17/52.9%/+6.25 -> data 22/27.3%/-15.43 ### Numeric claims checked
- lambda 0.35 cal_gap early: draft=-0.009, data=-0.0088959764298381
- lambda 0.35 cal_gap middle: draft=0.014, data=0.0141401032850679
- lambda 0.35 cal_gap late: draft=0.006, data=0.0056688569481091
- convective late edge null: draft=0.032, data=0.0324999999999999
- convective late edge 0.5: draft=0.045, data=0.0447926829268292
- convective late edge 0.4: draft=0.046, data=0.045890243902439
- D3 early final: draft=137.15, data=137.15
- D3 middle final: draft=142.93, data=142.925
- D3 late final: draft=110.01, data=110.005